Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18AS32]

# THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2021

## MEASUREMENT AND INSTRUMENTATION FOR AEROSPACE

**Time: 3 Hours** **Maximum Marks: 70**

**Instructions:**
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

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| **1.** | **a.** | What are basic flight instruments, air data instruments, navigational instruments and briefly explain their role. | **07** |
| | **b.** | What is 'basic T' arrangement in grouping cockpit instruments? Explain. | **07** |
| **2.** | **a.** | Explain what is Mach No. and what is its significance in Aviation? | **07** |
| | **b.** | Draw a schematic diagram explaining the operation of Pressure Altimeter. | **07** |
| **3.** | **a.** | What are the cockpit instruments that work on the principle of Gyroscope? Explain briefly their operation. | **07** |
| | **b.** | Give examples of Gyro which work with minimal or no mechanical moving parts. | **07** |
| | | **OR** | |
| **4.** | **a.** | What is a Gyroscope and what are its major applications in the field of aerospace? | **07** |
| | **b.** | Mention the errors of attitude indicators. | **07** |
| **5.** | **a.** | Explain the principle of operation of an engine speed measurement. | **07** |
| | **b.** | The Free turbine rpm of an engine in a helicopter is 10000. The signal output frequency is 667 HZ. What is the number of sensors on the shaft? | **07** |
| | | **OR** | |
| **6.** | **a.** | What is the basic principle of sensing rpm of a rotating device? Describe methods based on electrical and optical devices. | **07** |
| | **b.** | A shaft has been instrumented to generate 60 pulses per revolution. When the shaft runs at 4000 rpm compute the output frequency from the sensor. | **07** |
| **7.** | **a.** | Explain the following:<br>**i.** Absolute pressure<br>**ii.** Differential pressure<br>**iii.** Gauge pressure | **07** |
| | **b.** | How is fuel flow rate measured in aircraft /helicopter engines? | **07** |
| | | **OR** | |

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| **8.** | **a.** | Explain the operation of capacitance based fuel quantity measurement with a schematic diagram. | **07** |
| | **b.** | A turbine flow meter has a capacity to generate 512 pulses when a litre of fuel flows through it. What is the frequency of the output signal if the flow rate into the engine is 6 litres per minute? | **07** |